

Tentative Title: Predictive capacity of oceanographic indices of recruitment

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Abstract

To be determined...

0.1 General themes

- Balancing index stability versus predictive capacity. Generally, both are important for indices in stock assessment application. Due to the ephemeral nature of many indices of recruitment, and the challenges of changing ocean dynamics (nonstationarity, extremes on threshold or dome shaped relationships), its important that indices used to inform management decisions are both robust and can support predictions.
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0.2 If we only fit models from 1993 - 2010 do the inconsistencies in selection criteria change?

0.2.1 Methods

We repeated the same methods from the March analysis but for all species the models were only fit from 1993 - 2010. Covariates were re-standardized over the 1993 - 2010 period. We also omitted the rolling window analysis (as only two 15-year windows could be used with the reduced dataset).

0.2.2 Findings

This is not more consistent BUT the problem seems to shift. The issue here seems to be time series length rather than spanning a regime (training on 7 years of data and predicting 10 or training on 13 years of data and predicting 5 simply does not work well)

Overall, leave-future out criteria still performs poorly when applied to the 1993 - 2010 time period but this poor performance seems to be driven by short time series length rather than spanning seemingly distinct time periods. Additionally, the variables that are unstable from 1993 - 2018 tend to also be unstable over the 1993 - 2010 period. There seems to be two factors that jeopardize the utility of LFO in these circumstances. The first is whether or not there is an apparent regime shift in the oceanographic data and the second is time series length. Regardless of these two circumstances, variables that are unstable for a short time window without a regime shift seem to also be unstable in longer time series when a regime shift is included. Much like including the full time series, the best models, and the variables included in the best models, vary by selection criteria Table 1 - Table 4

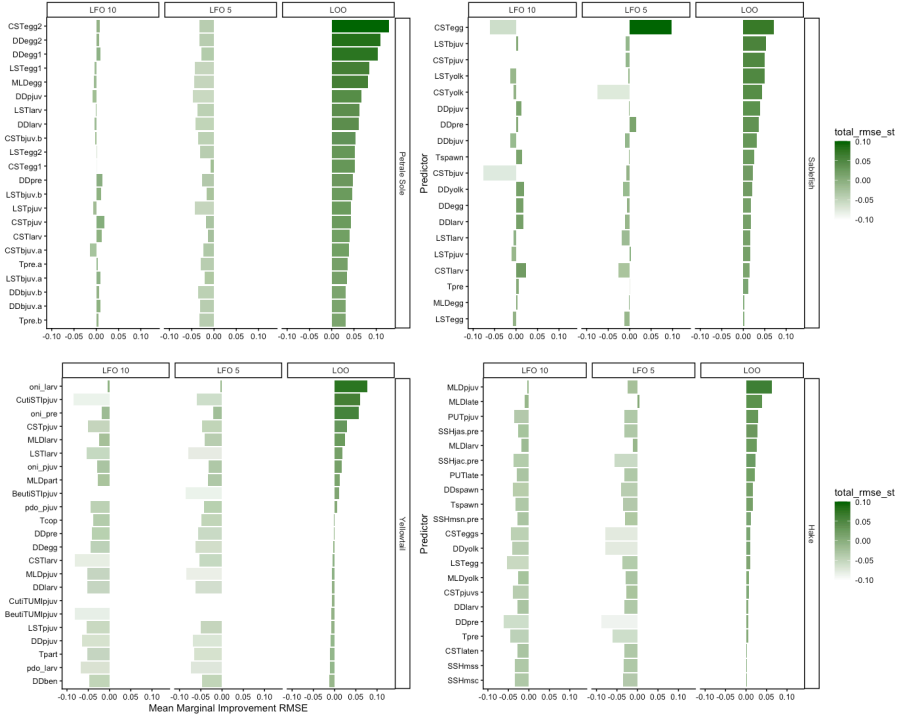


Figure 1: Marginal mean improvement of RMSE for each oceanographic driver and each selection criteria (leave-future-out last 10 years, LFO10; leave-future-out last 5 years, LFO5; leave-one-out all years)

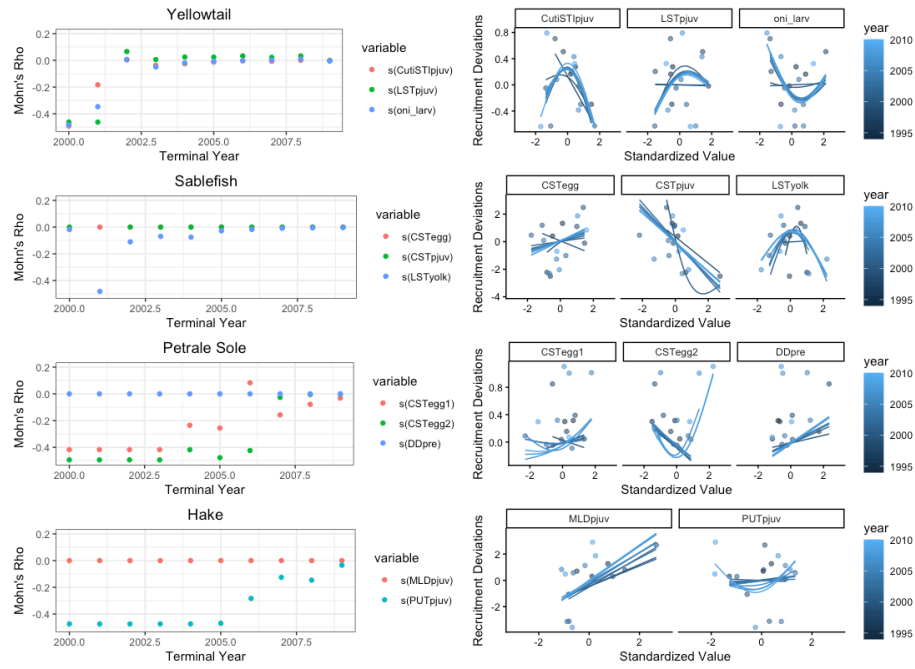


Figure 2: Mohn's rho (right) and associated functional relationships(left) for each of the 8 "peeled" years for each species. Mohn's rho can be interpreted as deviations from 0 indicate functional relationships that differ from the model fit to the full 1994 - 2010 time period

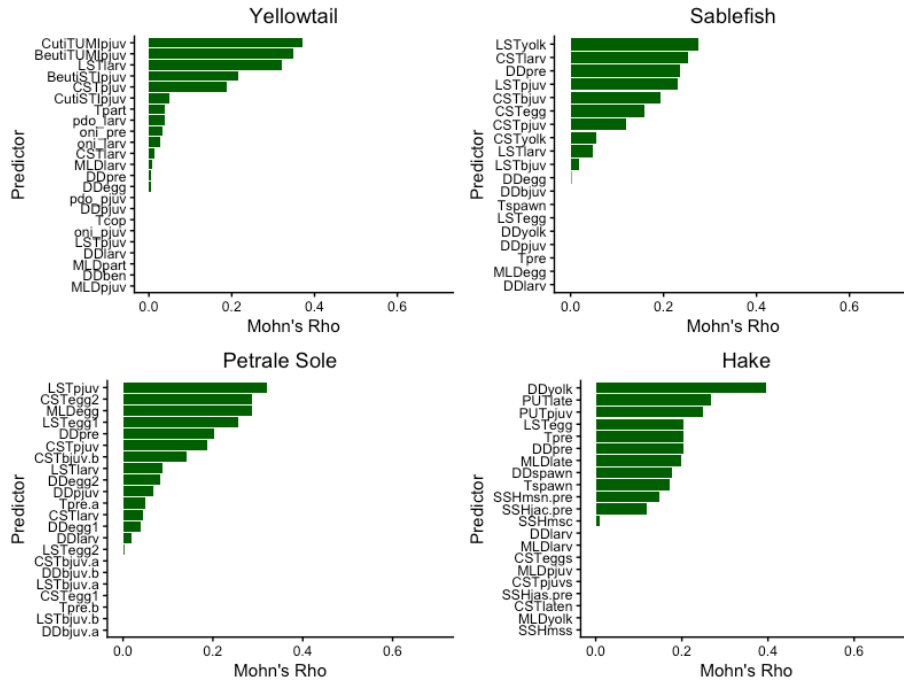


Figure 3: Absolute value of Mohn's rho for each oceanographic predictor for each of the 8 "peeled" years for each species. Mohn's rho can be interpreted as deviations from 0 indicate functional relationships that differ from the model fit to the full 1994 - 2010 time period

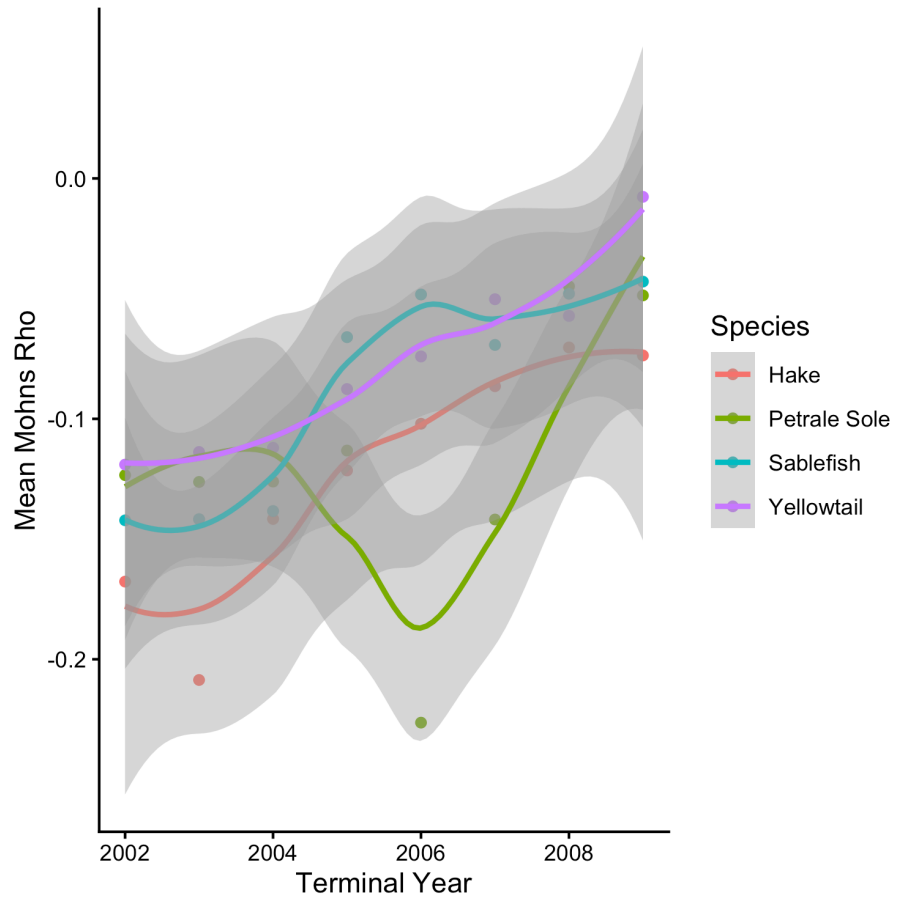


Figure 4: Mean Mohns rho by terminal year for single predictor models showing depicting mean values across oceanographic predictors by species with a loess smooth by species

Table 1: Top 5 models for petrale sole using each of the three selection criteria. Leave-future-out could not differentiate between 100 competing models

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CSTegg1 CSTegg2 DDpre	0.88	0.80	2.06	0.26	0.25	0.40	LOO
CSTbjuv.a CSTegg2	0.82	0.88	-5.91	0.27	0.30	0.42	LOO
LSTbjuv.b							
CSTegg2 DDpre	0.71	0.67	7.27	0.28	0.30	0.38	LOO
CSTegg2 DDpjuv DDpre	0.71	0.67	9.23	0.28	0.30	0.42	LOO
CSTegg2 CSTlarv DDpre	0.71	0.68	8.70	0.29	0.29	0.40	LOO
LFO5							
LSTegg2	0.82	0.11	20.45	0.44	0.25	0.40	LFO5
MLDegg Tpre.a	0.82	0.31	19.20	0.56	0.25	0.40	LFO5
LSTegg2 Tpre.b	0.82	0.10	22.47	0.47	0.25	0.40	LFO5
LSTbjuv.a MLDegg	0.82	0.28	19.90	0.50	0.25	0.40	LFO5
LSTbjuv.b LSTegg2	0.76	0.24	20.04	0.52	0.25	0.40	LFO5
LFO10							
CSTpjuv	0.88	0.10	20.72	0.42	0.30	0.38	LFO10
CSTegg2 DDpre	0.71	0.67	7.27	0.28	0.30	0.38	LFO10
CSTpjuv Tpre.a	0.88	0.17	21.67	0.44	0.30	0.38	LFO10
CSTbjuv.a Tpre.b	0.76	0.04	23.39	0.53	0.30	0.38	LFO10
CSTbjuv.b DDegg1	0.76	0.64	10.08	0.48	0.30	0.38	LFO10

Table 2: Top 5 models for sablefish using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CSTegg CSTpjuv LSTyolk	0.82	0.74	50.92	1.00	1.89	1.77	LOO
DDpjuv LSTyolk	0.76	0.62	55.67	1.09	1.90	1.83	LOO
CSTlarv DDpjuv LSTyolk	0.82	0.63	57.10	1.10	1.90	1.85	LOO
CSTegg DDpjuv LSTyolk	0.65	0.69	54.32	1.11	1.92	1.84	LOO
DDpjuv LSTyolk Tpre	0.76	0.63	57.18	1.12	1.92	1.84	LOO
LFO5							
LSTpjuv	0.29	0.07	67.06	1.79	1.63	1.50	LFO5
LSTpjuv Tpre	0.47	0.14	68.11	1.93	1.63	1.50	LFO5
CSTyolk Tspawn	0.53	0.34	64.64	1.59	1.63	1.50	LFO5
CSTegg LSTegg	0.65	0.35	64.40	1.65	1.63	1.50	LFO5
CSTegg LSTlarv	0.53	0.41	62.83	1.52	1.63	1.50	LFO5
LFO10							
LSTpjuv	0.29	0.07	67.06	1.79	1.63	1.50	LFO10
LSTpjuv Tpre	0.47	0.14	68.11	1.93	1.63	1.50	LFO10
CSTyolk Tspawn	0.53	0.34	64.64	1.59	1.63	1.50	LFO10
CSTegg LSTegg	0.65	0.35	64.40	1.65	1.63	1.50	LFO10
CSTegg LSTlarv	0.53	0.41	62.83	1.52	1.63	1.50	LFO10

Table 3: Top 5 models for yellowtail using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CutiSTIpjuv LSTpjuv oni_larv	0.82	0.84	5.38	0.30	0.33	0.34	LOO
CutiSTIpjuv LSTpjuv oni_pre	0.82	0.81	7.58	0.32	0.33	0.34	LOO
CutiSTIpjuv oni_larv	0.88	0.71	11.07	0.35	0.32	0.35	LOO
CutiSTIpjuv DDegg oni_larv	0.76	0.79	7.68	0.37	0.34	0.35	LOO
CSTpjuv CutiSTIpjuv oni_larv	0.82	0.71	12.75	0.37	0.34	0.44	LOO
LFO5							
oni_pjuv	0.53	0.15	23.65	0.47	0.29	0.31	LFO5
MLDpjuv Tpart	0.18	0.03	28.07	0.56	0.29	0.31	LFO5
MLDpart pdo_pjuv	0.53	0.24	24.09	0.52	0.29	0.31	LFO5
BeutiTUMIpjuv oni_pjuv	0.53	0.27	24.29	0.53	0.29	0.31	LFO5
BeutiSTIpjuv DDben DDpre	0.47	0.12	28.61	0.57	0.29	0.31	LFO5
LFO10							
oni_pjuv	0.53	0.15	23.65	0.47	0.29	0.31	LFO10
MLDpjuv Tpart	0.18	0.03	28.07	0.56	0.29	0.31	LFO10
MLDpart pdo_pjuv	0.53	0.24	24.09	0.52	0.29	0.31	LFO10
BeutiTUMIpjuv oni_pjuv	0.53	0.27	24.29	0.53	0.29	0.31	LFO10
BeutiSTIpjuv DDben DDpre	0.47	0.12	28.61	0.57	0.29	0.31	LFO10

Table 4: Top 5 models for hake using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
MLDpjuv PUTpjuv	0.88	0.61	62.82	1.44	2.33	2.11	LOO
PUTpjuv SSHjas.pre Tspawn	0.88	0.73	60.30	1.45	2.42	2.43	LOO
CSTlaten MLDpjuv PUTpjuv	0.88	0.61	64.62	1.48	2.39	2.34	LOO
MLDpjuv PUTpjuv SSHmsc	0.88	0.62	64.25	1.49	2.33	2.11	LOO
MLDpjuv PUTpjuv SSHmsn.pre	0.88	0.61	64.82	1.53	2.33	2.11	LOO
LFO5							
MLDlate	0.53	0.21	70.84	1.82	2.04	2.17	LFO5
MLDlate SSHjas.pre	0.47	0.30	70.99	1.82	2.04	2.17	LFO5
MLDlate SSHmsn.pre	0.47	0.23	72.56	1.97	2.04	2.17	LFO5
MLDlate SSHmsc	0.47	0.22	72.80	1.95	2.04	2.17	LFO5
MLDlate SSHmss	0.47	0.22	72.63	1.98	2.04	2.17	LFO5
LFO10							
MLDpjuv	0.71	0.30	68.81	1.71	2.33	2.11	LFO10
MLDpjuv SSHjas.pre	0.76	0.35	70.02	1.86	2.33	2.11	LFO10
MLDpjuv SSHmsn.pre	0.71	0.30	70.80	1.82	2.33	2.11	LFO10
MLDpjuv SSHmsc	0.59	0.31	70.71	2.01	2.33	2.11	LFO10
MLDpjuv SSHmss	0.59	0.31	70.60	2.31	2.33	2.11	LFO10

0.3 If we are more restrictive in the correlation we allow across oceanographic variables, are our results more consistent across ways of expressing predictive capacity?

0.3.1 Methods

We repeated the same methods from the March analysis but for all species the models only used combinations of variables that had a pearson's correlation coefficient of 0.3 or less to reduce the number of covariates.

0.3.2 Findings

Overall this does not help. The issues we are encountering is not because of correlations across time series, at least not when models are restricted to only 3 covariates and a correlation coefficient of 0.5

As expected, reducing the correlation coefficient has no impact on the highest ranked variables based on marginal mean improvement of RMSE and nor does it impact the ranking of stability. This is expected because of the nature of the ways these variables are calculated (either using models for only one variable based on stability or by reducing the impact of redundant variables in the case of marginal mean improvement RMSE).

But, we do see subtle differences in the rolling window analysis and the highest ranked models. Overall, this seems to be an ineffective way to minimize issues with predictive capacity indicating that overfitting or redundancy in oceanographic time series does not seem to be the primary issue.

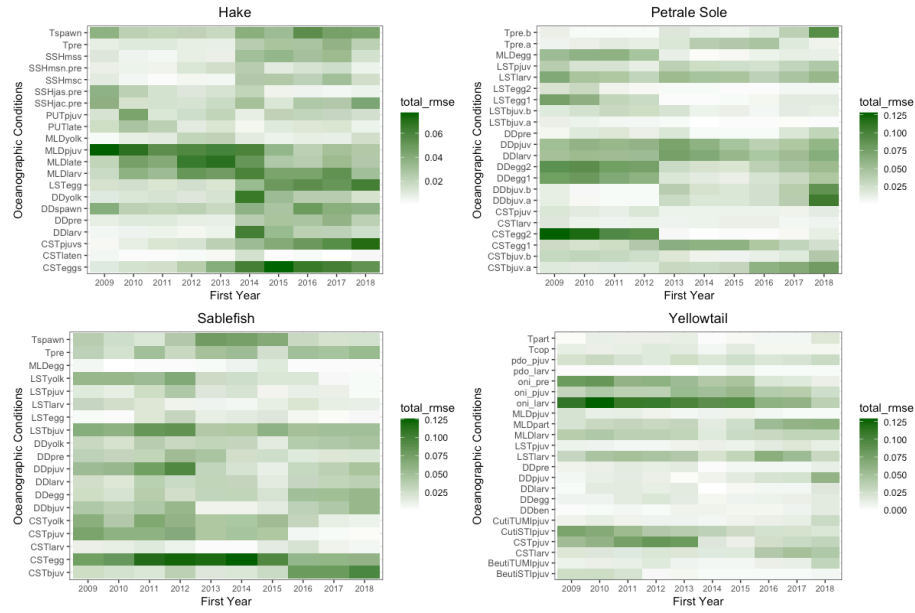


Figure 5: Marginal mean improvement of each oceanographic driver using 15-year time windows for each of the three groundfish species of interest

Table 5: Top 5 models for petrale sole using each of the three selection criteria. Leave-future-out could not differentiate between 100 competing models

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CSTbjuv.a DDpjuv LSTlarv	0.72	0.72	9.53	0.37	0.64	0.48	LOO
CSTbjuv.a DDlarv LSTlarv	0.68	0.74	8.18	0.37	0.62	0.51	LOO
CSTbjuv.a DDlarv	0.68	0.42	22.52	0.40	0.65	0.49	LOO
CSTbjuv.a DDpjuv	0.72	0.45	21.49	0.40	0.66	0.50	LOO
CSTbjuv.b DDlarv LSTlarv	0.64	0.57	18.85	0.40	0.63	0.49	LOO
LFO5							
LSTbjuv.b	0.72	0.01	32.16	0.44	0.61	0.48	LFO5
CSTpjuv DDegg2	0.72	0.19	29.75	0.48	0.61	0.48	LFO5
CSTegg1 CSTlarv	0.68	0.12	31.14	0.42	0.61	0.48	LFO5
CSTegg1 DDpre LSTlarv	0.64	0.27	30.16	0.51	0.61	0.48	LFO5
CSTegg1 CSTpjuv DDpre	0.64	0.20	30.82	0.51	0.61	0.48	LFO5
LFO10							
CSTpjuv	0.72	0.14	28.56	0.41	0.64	0.48	LFO10
DDbjuv.b Tpre.a	0.64	0.11	32.12	0.48	0.64	0.48	LFO10
CSTbjuv.a Tpre.b	0.72	0.23	28.26	0.44	0.64	0.48	LFO10
DDlarv LSTpjuv	0.60	0.42	24.40	0.47	0.64	0.48	LFO10
DDpjuv LSTpjuv	0.60	0.39	25.37	0.48	0.64	0.48	LFO10

Table 6: Top 5 models for sablefish using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CSTegg LSTbjuv	0.60	0.54	80.62	1.20	1.69	1.43	LOO
CSTegg DDlarv LSTbjuv	0.60	0.54	82.38	1.26	1.06	1.85	LOO
CSTegg LSTbjuv LSTlarv	0.56	0.55	82.00	1.26	1.63	1.57	LOO
CSTegg DDyolk LSTbjuv	0.60	0.54	82.22	1.26	1.56	1.84	LOO
CSTegg LSTbjuv LSTegg	0.60	0.54	82.42	1.27	1.06	1.85	LOO
LFO5							
CSTegg	0.48	0.18	90.61	1.44	1.06	1.85	LFO5
CSTlarv DDpre	0.40	0.10	93.45	1.54	1.06	1.85	LFO5
DDegg LSTlarv	0.32	0.06	95.05	1.59	1.06	1.85	LFO5
DDpjuv Tspawn	0.56	0.07	94.23	2.02	1.06	1.85	LFO5
DDyolk LSTbjuv	0.60	0.30	88.90	1.53	1.06	1.85	LFO5
LFO10							
Tspawn	0.44	0.03	93.31	1.53	1.55	1.33	LFO10
DDyolk	0.28	0.07	93.36	1.59	1.66	1.35	LFO10
LSTlarv Tspawn	0.56	0.07	94.37	1.56	1.66	1.35	LFO10
DDlarv LSTyolk	0.28	0.02	95.61	1.63	1.66	1.35	LFO10
DDlarv LSTpjuv	0.56	0.10	93.39	1.55	1.66	1.35	LFO10

Table 7: Top 5 models for yellowtail using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CutiSTIpjuv DDegg oni_pjuv	0.60	0.46	22.41	0.38	0.38	0.42	LOO
MLDlarv	0.60	0.15	26.43	0.38	0.33	0.33	LOO
LSTpjuv oni_larv	0.68	0.32	24.60	0.39	0.32	0.35	LOO
MLDlarv MLDpjuv	0.56	0.22	26.31	0.39	0.33	0.33	LOO
LSTpjuv MLDlarv	0.56	0.18	27.51	0.40	0.32	0.35	LOO
LFO5							
oni_pjuv	0.48	0.14	26.67	0.40	0.29	0.31	LFO5
oni_pjuv Tcop	0.56	0.18	27.56	0.41	0.29	0.31	LFO5
oni_pjuv Tpart	0.48	0.15	28.42	0.42	0.29	0.31	LFO5
oni_pjuv oni_pre	0.52	0.22	27.60	0.43	0.29	0.31	LFO5
MLDpjuv	0.44	0.03	29.78	0.42	0.32	0.38	LFO5
LFO10							
oni_pjuv	0.48	0.14	26.67	0.40	0.29	0.31	LFO10
oni_pjuv Tcop	0.56	0.18	27.56	0.41	0.29	0.31	LFO10
oni_pjuv Tpart	0.48	0.15	28.42	0.42	0.29	0.31	LFO10
oni_pjuv oni_pre	0.52	0.22	27.60	0.43	0.29	0.31	LFO10
MLDpart	0.64	0.11	27.59	0.40	0.35	0.32	LFO10

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Table 8: Top 5 models for hake using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_loo	RMSE5	RMSE10	SelectionCriteria
LOO							
CSTeggs CSTpjuvs MLDlate	0.68	0.48	97.23	1.72	2.33	2.11	LOO
CSTeggs CSTpjuvs MLDlarv	0.64	0.47	97.90	1.73	2.41	2.50	LOO
CSTeggs CSTpjuvs MLDyolk	0.68	0.46	98.36	1.73	2.33	2.11	LOO
CSTeggs CSTpjuvs MLDpjuv	0.56	0.45	98.67	1.79	2.38	2.42	LOO
MLDlate PUTpjuv	0.40	0.15	103.74	1.81	2.73	2.40	LOO
LFO5							
MLDlate	0.32	0.13	102.95	1.85	2.04	2.17	LFO5
DDyolk SSHmsc	0.40	0.08	106.21	2.10	2.04	2.17	LFO5
CSTlaten SSHjac.pre	0.40	0.14	105.47	1.99	2.04	2.17	LFO5
SSHmss Tpre	0.20	0.12	106.19	2.24	2.04	2.17	LFO5
DDpre LSTegg	0.28	0.09	106.66	2.26	2.04	2.17	LFO5
LFO10							
MLDpjuv	0.56	0.09	103.59	1.88	2.33	2.11	LFO10
PUTpjuv SSHmsc	0.40	0.10	106.18	2.14	2.33	2.11	LFO10
CSTeggs SSHmsc	0.44	0.12	106.02	2.27	2.33	2.11	LFO10
DDspawn SSHmss	0.36	0.06	107.14	2.29	2.33	2.11	LFO10
LSTegg SSHjac.pre	0.24	0.08	106.85	2.25	2.33	2.11	LFO10

0.4 If we remove “unstable” variables, is predictive capacity more consistent across variables?

0.4.1 Methods

To do this analysis we fit the analysis approach we used in March (full time series 1994 - 2018; correlation coefficient of 0.5) and removed all variables that had a Mohn’s rho of 0.1 (see Figure 3 for reference). This reduced the number of variables used for petrale sole by 7, for hake by 9, for yellowtail by 4, and for sablefish by 8.

0.4.2 Findings

Overall this does seem to produce clearer “winners” and “losers” than using all of the variables but selection criteria are still not consistent (with the exception of hake). There may be certain “types” variables that are inherently less stable than others (i.e., degree days)

Interestingly, yellowtail had the most stable best model in all analyses but also had the least number of unstable predictors. We hypothesized that this may be due to yellowtail having basin-scale time series (ONI and CUTI) as covariates but it also seems that overall there was just more stability across all of the time series. One other possible mechanism here is that yellowtail seems to have the smallest spatial range over which these covariates are summarized (exclusively within the NCC). Maybe characterizing them over such large spatial ranges contributes to instability?

When unstable variables are excluded, it becomes much more obvious which variables both provide stability (highly ranked in marginal mean improvement RMSE for LOO) and which also provide improved predictive capacity on 5 and 10 year time horizons (MMI RMSE for leave-future-out). Of the 4 species, there is at least 1 highly ranked LOO covariate that also provides improved predictive capacity (Figure 6).

There are also some general patterns of which types of variables that were considered for more than 1 species are unstable. For example, degree day variables are unstable 40% of the time when considered as a covariate of recruitment. **(NOTE TO SELF add in basin-scale indices once I add them to hake, consider months of year?)**

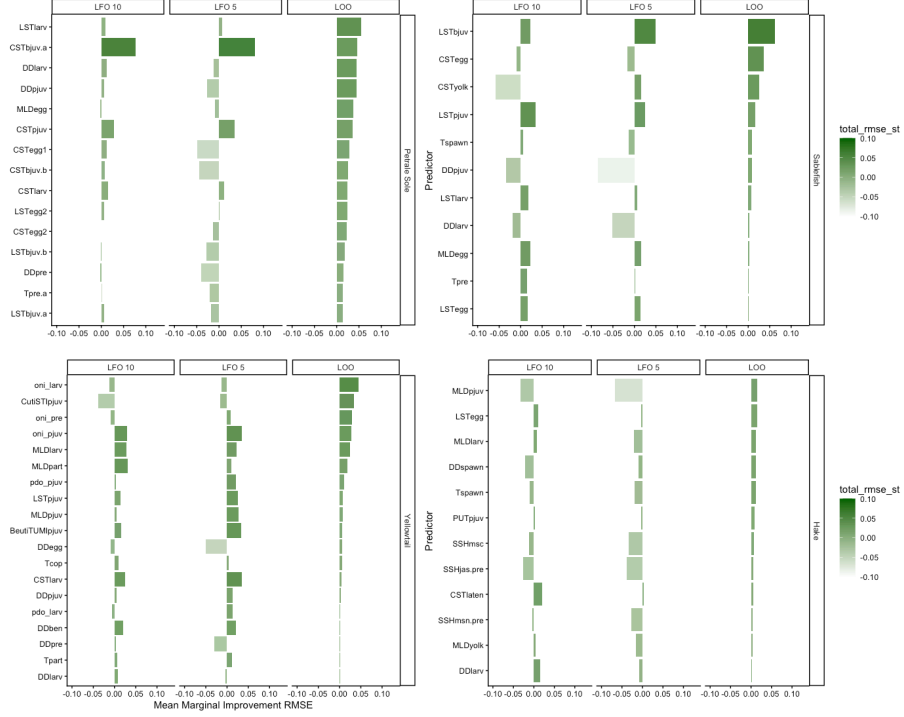


Figure 6: Marginal mean improvement of RMSE for each oceanographic driver and each selection criteria (leave-future-out last 10 years, LFO10; leave-future-out last 5 years, LFO5; leave-one-out all years)

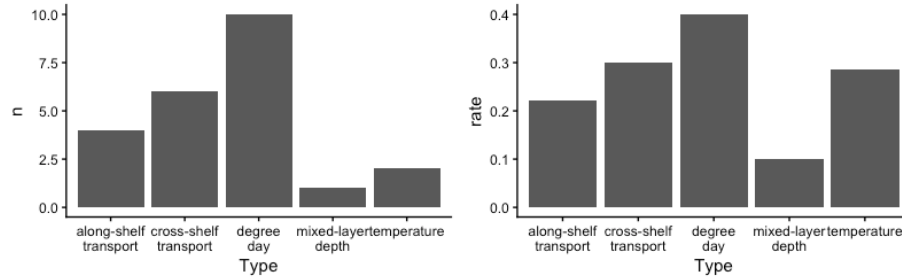


Figure 7: For types of covariates considered for more than one species, the total number (n) of covariates that were considered unstable and the rate (number unstable / total number considered)

0.5 If we used a “combined” selection criteria or “ensembled” best models, what do the predictions look like?

0.5.1 Methods

To consider which indicators had the best predictive capacity overall across metrics, marginal mean improvement was averaged across the 3 selection criteria.

To conduct this analysis, I took the results for the March analysis (1994 - 2018; correlation = 0.5). For every model, RMSE LOO and RMSE LFO 5-year were combined (summed) and RMSE LOO was given a weight of 2 to produce a “combined” RMSE. The model with the lowest combined RMSE was ranked as the best performing model. We then compared the 1-year ahead prediction of the last 5 years of this model to the best model with the other 3 selection criteria (LOO, LFO-5, LFO-10).

To consider an ensemble model, the predictions for 1-year ahead of the top 5 models for each of the 3 selection criteria (LOO, LFO-5, LFO-10) were averaged. No weighting was done.

0.5.2 Findings

- Combining selection criteria is informative for understanding which variables balance both predictive capacity and stability (Figure 10)
- In all circumstances, the best model using a combined selection criteria was one of the other highly ranked models, and helps distinguish between models with very similar RMSE for LFO (Figure 8)
- An ensemble approach isn’t really helpful, most interannual variability is washed out ()

Table 9: Top 5 models for yellowtail using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_combined	SelectionCriteria
Combined					
oni_pjuv	0.48	0.14	26.67	1.71	Combined
oni_pjuv Tcop	0.56	0.18	27.56	1.73	Combined
oni_pjuv pdo_larv	0.52	0.17	27.84	1.73	Combined
MLDlarv	0.60	0.15	26.43	1.75	Combined
oni_pjuv Tpart	0.48	0.15	28.42	1.75	Combined

Table 10: Top 5 models for petrale sole using each of the three selection criteria

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variables	Hit	dev.ex	AIC	RMSE_combined	SelectionCriteria
Combined					
CSTbjuv.a	0.76	0.16	27.96	1.57	Combined
CSTpjuv	0.72	0.14	28.56	1.64	Combined
CSTlarv	0.68	0.07	30.60	1.73	Combined
Tpre.b	0.64	0.08	30.38	1.74	Combined
LSTlarv	0.68	0.23	27.69	1.74	Combined

Table 11: Top 5 models for sablefish using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_combined	SelectionCriteria
Combined					
CSTegg LSTbjuv MLDegg	0.62	0.56	79.37	3.96	Combined
CSTegg LSTbjuv	0.62	0.54	78.43	3.96	Combined
LSTbjuv	0.58	0.30	84.77	3.98	Combined
CSTbjuv CSTegg LSTbjuv	0.67	0.63	75.34	4.04	Combined
LSTbjuv Tpre	0.54	0.31	86.42	4.05	Combined

Table 12: Top 5 models for hake using each of the three selection criteria

variables	Hit	dev.ex	AIC	RMSE_combined	SelectionCriteria
Combined					
CSTeggs CSTpjuvs MLDlate	0.68	0.48	97.23	7.31	Combined
CSTeggs CSTpjuvs DDspawn	0.68	0.56	94.99	7.32	Combined
CSTeggs CSTpjuvs MLDlarv	0.64	0.47	97.90	7.34	Combined
CSTeggs CSTpjuvs MLDyolk	0.68	0.46	98.36	7.34	Combined
CSTeggs CSTpjuvs Tspawn	0.68	0.54	96.02	7.42	Combined

0.6 Are basin scale variables more stable than these local variables?

0.6.1 Methods

Procure data from KM and NT Add Hake and sablefish variables as listed in the notes

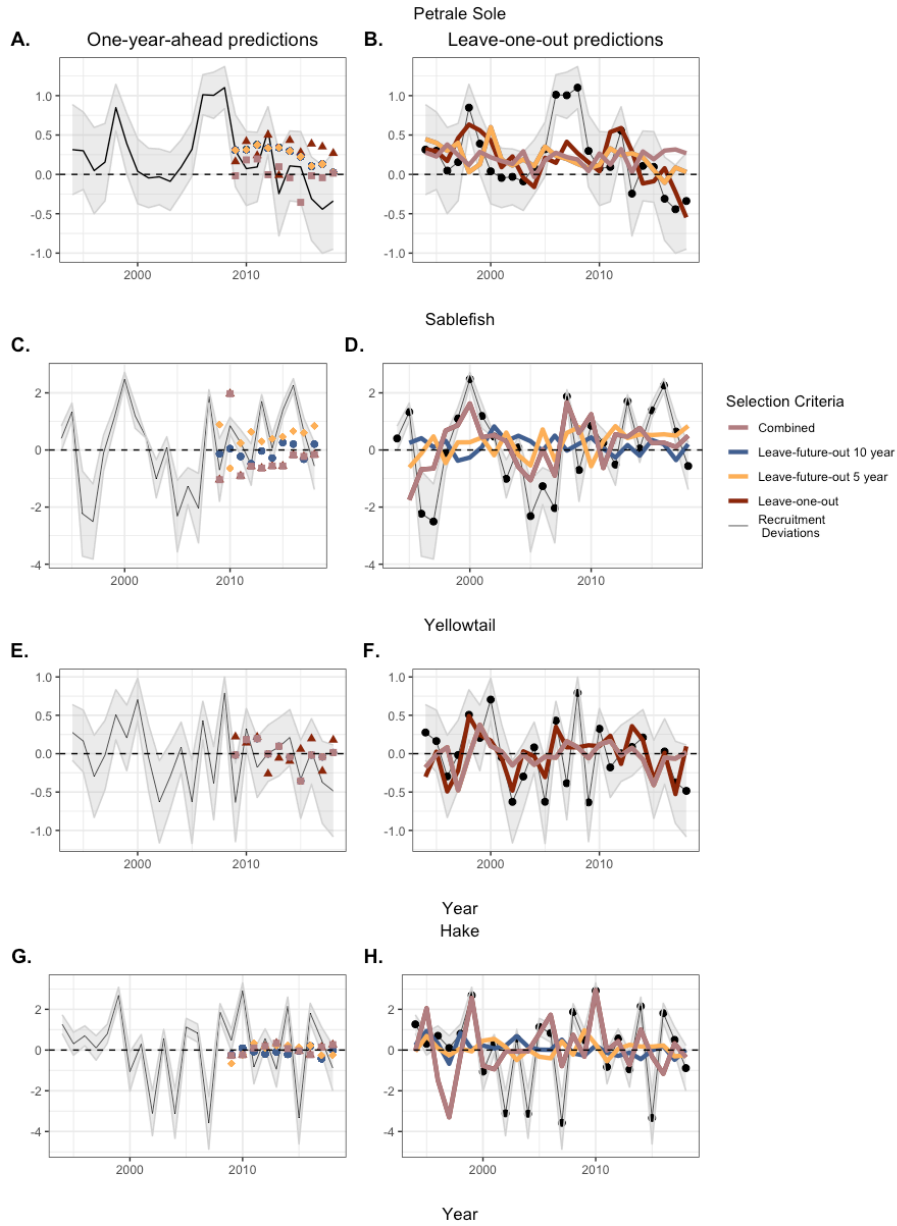


Figure 8: One year ahead predictions of recruitment deviations (black line) for the last ten years (A., B., C.) and leave-one-out predictions for all years (B., D., F.) for the best supported models based on RMSE for one year ahead predictions for the last 10 years (LFO10, blue), last 5 years (LFO5, yellow), RMSE for leave-one-out predictions (red, LOO) and combined selection criteria (pink). Rows depict individual species, Petrale Sole (A.,B.), Sablefish (C. D.) and Yellowtail rockfish (E. F.)

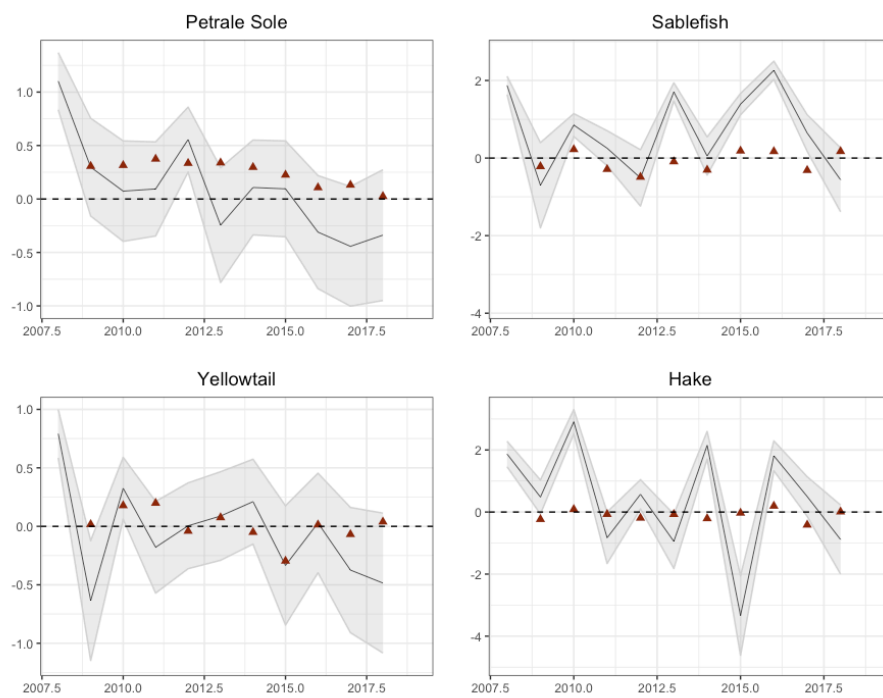


Figure 9: One-year-ahead predictions for the ensembled models

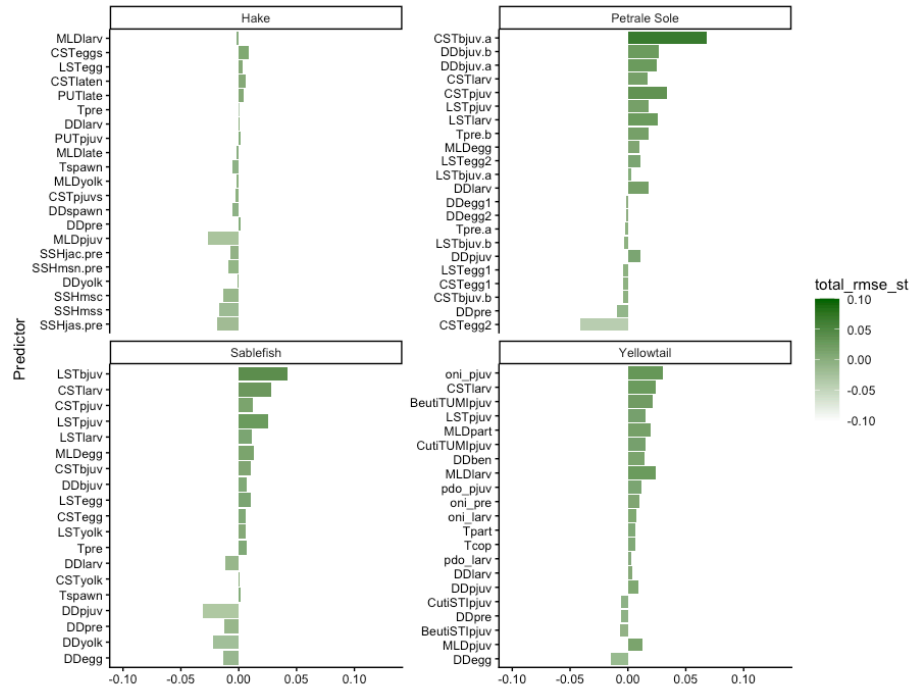


Figure 10: Marginal mean improvement of RMSE for each oceanographic driver across selection criteria (leave-future-out last 10 years, LFO10; leave-future-out last 5 years, LFO5; leave-one-out all years)

1 TO DO

1.1 Does seasonality impact stability?

1.2 Are there patterns in the relative importance of variables through time?

1.2.1 Methods

DFA on rolling window